DURGA NIRMALESWARAN

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Union City, NJ, USA

EDUCATION

University of Massachusetts Amherst *Aug 2024 - May 2026*
M.S. in Computer Science | GPA: 3.8/4.0

Anna University, India *Aug 2020 - Apr 2024*
B.Tech. in Information Technology | CGPA: 8.7/10 **(Top 10% • Merit Scholarship Recipient)**

SKILLS

Languages	Python, C/C++, HTML/CSS, JavaScript, R, SQL
Data Visualization	D3.js, Matplotlib, Quarto, RStudio, SVG
Web & Tools	Node.js, Git, GitHub, Linux, Jupyter Notebook, REST APIs
Bioinformatics	BLAST, PSI-BLAST, DIAMOND, k-mer Features, Protein Sequence Analysis
Data Engineering & Analysis	NumPy, Pandas, Apache Flink, Feature Engineering, Hypothesis Testing, Regression Analysis
AI/ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, CNNs, Transformers, Attention Mechanisms, Computer Vision, Reinforcement Learning
NLP & LLMs	Hugging Face Transformers, Sentence-BERT, RoBERTa, FAISS, NLTK, VADER, RAG, LLM Prompting, OpenAI API, Multimodal LLM

WORK EXPERIENCE

Research Assistant @ *Human Computer Interaction and Visualization* Lab , *UMass Amherst* *June 2025 - Present*

- Built an AI pipeline that reads raw SVG chart files and automatically identifies what each visual element is - axes, data marks, gridlines - achieving 82-86% accuracy across 17,276 elements in 102 real-world charts.
- Ran an ablation study showing that removing cohort-based decomposition tanked accuracy by 65+ percentage points, proving it's the core reason the pipeline works.

Research Assistant @ *Socially Intelligent Media and Systems* Lab, *UMass & University College Dublin* *June 2025 - Present*

- Built a pipeline to collect 1M+ X/Twitter posts linked to scientific papers via Altmetric and normalize inconsistent API responses into a consistent canonical format for downstream NLP analysis.
- Analyzed CS & Physics related tweet datasets to compare hype-word usage across disciplines and found that CS papers had noticeably higher hype rates than Physics, suggesting social media visibility may inflate perceived scientific impact.

Teaching Assistant, Machine Learning @ *UMass Amherst* *Jan - May 2026*

- Supported 225+ grad/undergrad students in applying core ML concepts through structured problem-solving guidance, debugging support, and detailed technical feedback.
- Reviewed submissions for reproducibility, required outputs, and technical correctness while coordinating fair evaluation and regrade workflows with course staff.

Graduate Student Intern @ UMass Amherst (College of Humanities & Fine Arts) *Jan - May 2026*

- Researched how AI infrastructure contributes to environmental and material costs as part of a cross-disciplinary initiative rethinking sustainability at scale, turning findings into actionable timelines and deliverables.
- Mentored an undergrad researcher and partnered with faculty PIs to keep research on track, refine direction, and make sure outcomes landed clearly across the team.

Web Developer Intern @ **Ideassions Technology Solutions**, India *Aug 2023*

- Built and scaled a responsive hotel booking web app with a 4-person team, developing front-end components that streamlined the booking flow and improved usability using HTML/CSS & JS.

NOTABLE PAPERS

- *"Cohort-based Semantic Labeling: AI-Enabled Recovery of Visualization Semantics from Deployed SVGs"* - Submitted to **IEEE VIS**; an AI-based approach to **recover semantic meaning from SVG charts**, enabling richer interpretation and analysis.
- *"Sentiment Analysis of Amazon Reviews"* - Published in the ***International Journal of Innovative Research in Computer and Communication Engineering (IJIIRCCE)***, Vol. 12, Issue 5, May 2024. Explored the integration of **NLTK and transformer-based models** for sentiment classification, providing insights into customer opinion mining.
- *"Visual Image Captioner using Deep Learning"* - Presented at the **International Conference on Intelligent Computing, Advanced Communication and Materials *(IC2ACM-2023)***. Developed a **CNN-RNN-based image captioning system**.

TECHNICAL PROJECTS

ENZYME CLASSIFICATION BENCHMARKING

- Compared ML and bioinformatics baselines for predicting enzyme EC classes from amino-acid sequences using the DGEB benchmark and implemented logistic regression, Random Forest, k-mer, prototype, BLAST, PSI-BLAST, and DIAMOND baselines, evaluating performance with weighted F1.
- Found that BLAST reached 0.56 weighted F1 in 3.3 seconds on CPU, outperforming multiple protein foundation-model baselines with much lower compute.

RAG-BASED CLAIM VERIFICATION SYSTEM

- Built the few-shot ICL classifier for a RAG-based fake news detection system, classifying claims into 4 veracity labels by grounding LLM predictions in retrieved evidence rather than model memory alone.
- Designed the evaluation framework integrating METEOR and Averitec scoring to measure pipeline quality across multiple LLMs.

AUTOMATED IMAGE DESCRIPTION MODEL

- Developed a model using EfficientNet B7 for visual feature extraction and a Transformer decoder with self-attention to generate contextually accurate captions from images.
- Trained on the Flickr dataset with image augmentation and positional encodings, enabling the model to understand spatial relationships between objects and produce human-readable descriptions.

AI-DRIVEN AMAZON REVIEW SENTIMENT ANALYSIS

- Analyzed ~500K Amazon food reviews using VADER and RoBERTa to classify customer sentiments, finding RoBERTa gave sharper positive/negative separation while VADER was faster on large data.
- Compared Logistic Regression, BernoulliNB, and LinearSVC classifiers - Logistic Regression came out on top at 81% accuracy, while BernoulliNB hit 79% with significantly faster training time.

OTHER NOTABLE PROJECTS

E-COMMERCE ANALYTICS WITH RSTUDIO – GLOBAL SUPERSTORE

- Analyzed 65K+ retail transactions in RStudio to uncover sales and profitability trends through data cleaning, visualization, and exploratory modeling.
- Built a reproducible Quarto analytics report with interactive visuals that made findings easy to communicate and verify.

INTERACTIVE DASHBOARD FOR US TRAFFIC ACCIDENT TRENDS

- Built an interactive D3.js dashboard to explore 100K+ US traffic accident records across state, severity, and road infrastructure dimensions with animated transitions, hover tooltips & tab-based navigation for smooth data exploration.
- Found that traffic signals accounted for the highest accident volumes while roundabouts had significantly fewer incidents, with California leading all states at 256K+ accidents from 2016-2023.

COMPARATIVE STUDY OF REINFORCEMENT LEARNING METHODS

- Built three RL algorithms from scratch - One-Step Actor-Critic, True Online SARSA(λ) & Prioritized Sweeping and ran 10 experiments per algorithm across two stochastic grid environments to see how they actually compare.
- Prioritized Sweeping was the clear winner in speed, converging 6x faster than model-free methods, while Actor-Critic ended up with the best final accuracy (MSE 0.094) after tuning 12 hyperparameter configurations per algorithm.

ML MODEL COMPARISON PROJECT

- Implemented k-NN, Random Forest, and Neural Networks completely from scratch in Python and benchmarked them across 4 datasets - MNIST, Parkinson's, Rice, and Credit Approval using stratified 10-fold cross-validation.
- Found that Random Forest hit 97.57% accuracy on MNIST while simpler models like k-NN outperformed Neural Networks on smaller datasets, showing that more complex doesn't always mean better.

RIDESHARE FARE PREDICTION & PRICING ANALYSIS

- Analyzed 637K+ Uber and Lyft ride quotes from Boston using t-tests, ANOVA, and regression to identify what actually drives fares - finding that weather and time of day had almost no effect while service tier and distance dominated.
- Built a fare prediction model using just 5 variables - service tier, distance, surge multiplier, and route that hit R^2 of 0.9286 and predicted fares within $2.49 on held-out test data.